



C20-EE-504

7648

BOARD DIPLOMA EXAMINATION, (C-20)

MARCH/APRIL—2025

DEEE – FIFTH SEMESTER EXAMINATION

POWER ELECTRONICS, PLC AND SCADA

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. State any three thyristor family devices.
2. Draw the gate characteristics of SCR.
3. Define the terms (a) AC regulator and (b) chopper.
4. Draw the circuit of three-phase fully controlled converter with resistive load.
5. Classify inverters based on (a) type of output voltage and (b) type of connection.
6. List any three applications of cyclo-converters.
7. List the factors affecting the speed of an AC induction motor.
8. State any three devices used to suppress the spikes in supply system.
9. State any three applications of PLCs.
10. State any three softwares used in SCADA.

PART—B

8×5=40

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **eight** marks.
(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. (a) Explain the SCR circuit triggered by UJT with neat circuit diagram.

(OR)

(b) Explain the volt-ampere characteristics of DIAC under forward and reverse bias.

12. (a) Explain the working of single-phase half wave controlled converter under R-L load with neat waveforms.

(OR)

(b) Explain the operation of chopper in all its four quadrants.

13. (a) Draw the circuit diagram of parallel inverter and explain its operation.

(OR)

(b) Define cyclo-converter and also explain the working principle of cyclo-converter.

14. (a) Explain the speed control of DC shunt motor by using chopper with neat circuit diagram.

(OR)

(b) Explain the operation of battery charging circuit using SCR with the help of a neat circuit diagram.

15. (a) Draw the block diagram of PLC and explain about the purpose of each part.

(OR)

(b) Draw the ladder diagram for traffic light controller and explain the operation with a neat diagram.

PART—C

10×1=10

- Instructions :**
- (1) Answer the following question.
 - (2) The question carries **ten** marks.
 - (3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

16. State the need of SCADA and explain the working of SCADA with PLC.

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